

Anytime-valid monitoring of repeated survey estimates under questionnaire redesign: An application to the Business Trends and Outlook Survey

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Abstract

Statistical agencies release survey estimates every few weeks, and users read each release as it arrives, asking whether growth has changed course. Repeated significance testing does not control error rates on such a stream, and questionnaire redesigns create discontinuities that are easily mistaken for real change. We propose a monitor for the curvature of a published series: a mixture e-process, valid under continuous inspection and optional stopping, for the null that the latent path is locally linear with unknown level and slope and a separate unknown level for each questionnaire regime. Extensions cover an unknown common scale of the published standard errors, proportional redesign effects, regime dates that are off by one wave and the choice of prior scale. Simulations show where each version holds its level and what it costs; dependent errors are not covered. Applied to 56 waves of the U.S. Census Bureau's Business Trends and Outlook Survey on artificial intelligence use, the recommended monitor declares a deceleration in April 2024, triggered by one large drop, and accelerations in October 2024 and March 2025. No declaration remains if that drop is a level shift, and a state-space model with a random-walk level and constant growth fits equally well.

Keywords

sequential monitoring, e-values, group invariance, survey redesign, repeated surveys, official statistics

1 Introduction

National statistical offices increasingly publish short-period estimates from rolling surveys. The U.S. Census Bureau's Business Trends and Outlook Survey (BTOS) releases national, sector, state and size-class estimates every two weeks (U.S. Census Bureau 2025). Users, from journalists to central bank staff, read each release against its predecessors and ask whether the path of an indicator has changed, for example whether the adoption of a new technology has accelerated or stalled (Bonney et al. 2024; Federal Reserve Board 2026; McElheran et al. 2024). Two features of this practice are not handled by standard inferential tools.

The first is continuous inspection. A user who applies a conventional test at every release, and who may stop at any time, faces an error rate that is not the nominal one (Armitage et al. 1969). The second is redesign. Agencies revise questions, weighting and frames, and the published series then contains a discontinuity that is not a feature of the population. The survey literature handles this with models for repeated surveys that include an intervention or a parallel run, estimated by state-space methods that can be run in real time (Scott and Smith 1974; van den Brakel and Roels 2010; O'Hara et al. 2019; Durbin and Koopman 2012). These methods estimate the size of the discontinuity and of the underlying slope, which is what an analyst usually wants, but they do not give finite-sample error control for a user who inspects the series after every release.

Sequential analysis (Wald 1945; Page 1954; Brown et al. 1975; Lan and DeMets 1983; Chu et al. 1996) and its modern form based on e-values and test martingales (Howard et al. 2021; Vovk and Wang 2021; Ramdas et al. 2023; Grünwald et al. 2024; Shin et al. 2023), including always-valid monitoring of online experiments (Johari et al. 2022), provide tests with error control that is uniform over time. We combine this with the removal of nuisance parameters by projection onto the residual space, in the spirit of invariance arguments for e-values (Pérez-Ortiz et al. 2024). The construction is a Robbins mixture martingale for one coefficient of a Gaussian regression with known variances; its novelty lies in the choice of the null (local linearity with a separate level for each questionnaire regime), in the treatment of regimes that enter the design sequentially, and in the evaluation on a real survey series, not in the martingale itself.

The contributions are as follows. First, a monitor for curvature of a published survey series with a closed form and a proof of validity for all values of the level, the slope and the regime offsets (Section 3). Second, extensions that remove the assumptions on which the basic monitor fails: unknown scale of the published standard errors, proportional redesign effects, regime dates that are off by one wave, and the choice of prior scale (Subsection 3.6). Third, a simulation study that reports where each version holds its level, what it costs in power, and where it still fails (Section 4). Fourth, an application to 56 BTOS waves in which we report how each declaration changes across the versions, and compare the monitor with a local-linear-trend state-space model that estimates the slope directly (Section 5).

2 Setting

Let $j = 1, 2, \dots$ index published waves, with release times t_j measured in two-week units and published estimates y_j with design-based standard errors s_j . Each wave carries a questionnaire regime label $r_j \in \{1, \dots, K\}$ that is fixed by the agency

before the estimate is released. We write $w_j = 1/s_j^2$. The model for the published estimate is

$$y_j = \theta(t_j) + \mu_{r_j} + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, s_j^2) \text{ independently,} \quad (1)$$

where θ is a smooth latent path and μ_r is the offset of regime r relative to the first regime. A redesign changes μ_r and leaves θ unchanged. The model is additive on the scale of the estimate: a redesign that rescales the series, or that changes its dynamics, is outside it. The standard errors s_j are treated as known and as fixed in advance, as is customary in the analysis of published time series; Subsections 4.1 and 5.2 examine what happens when they are too small. The normal approximation is reasonable at the sample sizes involved, and consecutive waves are drawn from different rotation panels (U.S. Census Bureau 2025), although a panel returns after twelve weeks and rotation group effects are a known source of bias in panel surveys (Bailar 1975).

The question asked of the series is about shape: over a stretch of consecutive waves, is the growth rate constant, or has it changed? For an epoch beginning at wave j_0 the null hypothesis is

$$H_0 : \quad \theta(t) = a + bt \quad \text{for some } a, b \in \mathbb{R}, \quad (2)$$

with a, b and the regime offsets μ_r unknown. The alternative is constant curvature, $\theta(t) = a + bt + \gamma t^2/2$, with $\gamma > 0$ for acceleration and $\gamma < 0$ for deceleration. A sustained change of the growth rate in one direction has a nonzero projection on the quadratic term and is detected as a departure from H_0 . Exact linearity is a null that no smooth path satisfies, so Subsection 3.1 also gives a version with a tolerance for small curvature.

3 Method

3.1 A mixture e-process for curvature

Fix an epoch and consider its first n waves. Let D_n be the $n \times p$ matrix whose columns are the indicators of the regimes present and the time variable t_j , let $W_n = \text{diag}(w_1, \dots, w_n)$, and let $q_j = t_j^2/2$. Define $M_n = I - D_n(D_n^\top W_n D_n)^{-1} D_n^\top W_n$ and

$$A_n = q^\top W_n M_n y, \quad B_n = q^\top W_n M_n q, \quad (3)$$

with q and y truncated to the first n waves. Here B_n is the information about γ after the level of each regime and the common slope have been removed by weighted least squares, and A_n/B_n is the corresponding estimate of γ , with standard error $B_n^{-1/2}$. Neither depends on how t_j is centred.

For a prior scale $\tau > 0$ on γ (in percentage points per squared wave) and a tolerance $\gamma_0 \geq 0$, the one-sided half-normal mixtures are

$$E_n^+ = \frac{2}{\tau \sqrt{a_n}} \Phi\left(\frac{A_n^+}{\sqrt{a_n}}\right) \exp\left(\frac{(A_n^+)^2}{2a_n}\right), \quad E_n^- = \frac{2}{\tau \sqrt{a_n}} \Phi\left(\frac{A_n^-}{\sqrt{a_n}}\right) \exp\left(\frac{(A_n^-)^2}{2a_n}\right), \quad (4)$$

where $a_n = B_n + \tau^{-2}$, $A_n^+ = A_n - \gamma_0 B_n$, $A_n^- = -A_n - \gamma_0 B_n$ and Φ is the standard normal distribution function. With $\gamma_0 = 0$ this is the closed form of $\int_0^\infty \exp(\gamma A_n - \gamma^2 B_n/2) \pi_\tau(d\gamma)$, with π_τ the half-normal law (Robbins 1970); for $\gamma_0 > 0$ the likelihood ratio is taken against the boundary value γ_0 , and the formula is the same with A_n shifted by $\gamma_0 B_n$. The statistic E_n^+ is evidence for acceleration and E_n^- for deceleration. Monitoring starts at the first wave n with $B_n > 0$, which is the third wave of an epoch whose waves share a regime.

Proposition 1. Let $\alpha \in (0, 1)$. Suppose model (1) holds with $\theta(t) = a + bt + \gamma t^2/2$ and with t_j, s_j and r_j known in advance, and $\gamma \leq \gamma_0$ (that is, H_0 when $\gamma_0 = 0$). Then for every a, b and regime offsets μ_r ,

$$\mathbb{P}(\exists n : E_n^+ \geq 2/\alpha) \leq \alpha/2,$$

and likewise for E^- when $\gamma \geq -\gamma_0$. The probability that either monitor ever reaches $2/\alpha$ is therefore at most α .

The proof is in Appendix A. The guarantee holds at all times simultaneously, so the user may inspect the series after every release and stop at any time. The declaration rule is the first time either monitor reaches $2/\alpha$ (40 for $\alpha = 0.05$). Below the threshold the e-values are not given verbal labels, because under the null the probability that E^+ or E^- ever exceeds a moderate level is not small: over 52 looks it is about 0.57 for a level of 2 and 0.09 for 10. As a summary of the evidence in an epoch without a declaration we report the anytime-valid p-value $\min\{1, 2/\max_n \max(E_n^+, E_n^-)\}$.

3.2 Redesigns as nuisance levels

The regime offsets μ_r enter D_n as separate columns. A new regime adds a column that is zero on all earlier rows, so the residual σ -fields used in the proof remain nested. Proposition 1 then holds for every value of the offsets, and a redesign that shifts the level by a constant of any size cannot produce a false declaration. Three conditions are needed and none is innocuous. The shift must be additive on the scale of the series; the post-redesign slope and curvature must be the same as before the redesign; and the regime label must be placed at the correct wave. Subsection 4.3 shows that a proportional effect, or a label that is off by one wave, destroys the guarantee of the basic monitor; Subsection 3.6 gives versions that do not need

these conditions, at a cost in information (Table 5). The size of the jump is not used, which costs information: a regime with one wave is absorbed completely, and a regime with two waves contributes one within-regime difference, so over a short post-redesign run the regime-aware monitor is equivalent to dropping the transition wave.

For comparison we also report the pooled analysis, in which the regime columns are dropped and the spliced series is treated as one regime.

3.3 Epochs and restarts

After a declaration the monitor starts a new epoch at the following wave, with fresh level and slope parameters.

Corollary 1. Suppose the latent path is linear on the epoch's span so far. For an epoch beginning at a stopping time, the conditional probability that it ends in a declaration is at most α . If the latent path is linear throughout, the number N of declarations satisfies $\mathbb{P}(N \geq m) \leq \alpha^m$ and $\mathbb{E}[N] \leq \alpha/(1 - \alpha)$.

This is the sense of error control for a sequence of declarations. It says that a declaration is rarely spurious when the path is linear. It says nothing when the path is not linear, which for any smooth adoption curve is the typical case: a declaration then records that the epoch's path has curved, and with the information B_n growing, small curvature is eventually declared. The tolerance γ_0 addresses this: with $\gamma_0 > 0$ the guarantee holds for every path whose curvature stays below γ_0 . A declaration is also a crossing time, not an estimate of when the change occurred, and the dates depend on where the epoch starts. We therefore report the estimated curvature A_n/B_n with its standard error at each declaration.

3.4 Cross-strata attribution check

When an agency announces a redesign the regime label is known. When it does not, the label may have to be inferred. Let $z_{k,j}$ be the standardized change in stratum k between waves $j - 1$ and j . An instrument event moves every stratum in the same wave, with the same sign and with magnitudes far above the stratum's own distribution of within-regime movements, whereas a change in the latent path builds up over several waves. We compare the change of each stratum with the 95th percentile of its own within-regime $|z_{k,j}|$, and count the strata beyond it with the same sign. This is a screening rule with no formal error control. We apply it to every wave and not only to announced redesigns.

3.5 Constants

The level is $\alpha = 0.05$, and each monitor is run at threshold $2/\alpha = 40$. The prior scale $\tau = 0.01$ percentage points per squared wave is a change of 0.1 points per wave in the growth rate accumulating over ten waves. This scale is not derived from a power calculation; the value was used in all analyses reported here, but other values were not excluded in advance, and the results for τ between 0.0025 and 0.04 are reported in Subsection 5.4. Validity does not depend on τ , only power and delay do, and Subsection 3.6 gives a mixture over a grid of prior scales that replaces the single value by an equal-weight average over five. The tolerance is $\gamma_0 = 0$ unless stated. Which monitor is primary matters for the application, where many variants are reported; we designate the unknown-scale monitor with the mixture over the prior scale as the recommended one and report it first in Subsection 5.3, and say there that the designation followed the exploratory analysis.

3.6 Extensions

The basic monitor needs four things to be right: the published standard errors, the additivity of redesign effects, the regime dates and the prior scale. This subsection gives a version that relaxes each. Their simulated behavior is in Subsection 4.4.

3.6.1 Unknown scale of the standard errors Suppose that the published standard errors are correct up to an unknown common factor, $\varepsilon_j \sim \mathcal{N}(0, \kappa^2 s_j^2)$ with $\kappa > 0$ unknown. The nuisance parameters are now the level, the slope, the regime offsets and κ . Let $\nu_n = n - \text{rank}(D_n)$ be the residual degrees of freedom, $\text{RSS}_n = y^\top W_n M_n y$, $z_n = A_n / \sqrt{\text{RSS}_n}$ and $J(c, \nu) = \int_0^\infty x^{\nu-1} e^{-x^2/2+cx} dx$. For a standardized curvature $\delta = \gamma/\kappa > 0$ define

$$L_n(\delta) = e^{-\delta^2 B_n/2} \frac{J(\delta z_n, \nu_n)}{J(0, \nu_n)}, \quad E_n^{\sigma,+} = \int_0^\infty L_n(\delta) \pi_\tau(d\delta), \quad (5)$$

with π_τ the half-normal law on δ , which has the same units as γ , and $E_n^{\sigma,-}$ with z_n replaced by $-z_n$. Monitoring starts when $\nu_n \geq 2$. The integral over δ has no closed form and is computed numerically on a grid that covers the prior up to six prior scales and the neighborhood of the maximum of the integrand; for $B_n \geq 10^3$ it agrees with exact integration to within 10^{-3} in mean (Appendix B).

Proposition 2. Suppose model (1) holds with $\theta(t) = a + bt + \gamma t^2/2$, $\gamma = 0$ and $\varepsilon_j \sim \mathcal{N}(0, \kappa^2 s_j^2)$. Then for every a, b, μ_τ and $\kappa > 0$, $(E_n^{\sigma,\pm})$ is a nonnegative martingale with mean one, and $\mathbb{P}(\exists n : E_n^{\sigma,\pm} \geq 2/\alpha) \leq \alpha/2$.

The proof is in Appendix A. The proposition covers the point null $\gamma = 0$ and not a tolerance. Its price is the information on the size of the errors in the first waves of each epoch and the numerical integral. Anytime-valid t -type constructions for unknown variance exist for simpler models (Wang and Ramdas 2025); the present one is built on the scale-invariant density in the same spirit as Pérez-Ortiz et al. (2024). It covers a common unknown factor only; weights that are wrong by wave-specific amounts, and dependent errors, are examined by simulation.

Table 1. Probability of at least one declaration over 52 looks (20,000 replications; nominal level 0.05). The first three rows are valid cases; correlated errors use a lag-6 autoregressive structure at the stated correlation, and standard errors are too small by the stated factor. Rows 4 to 10 use slope 0.15.

Scenario	E-process	Bonferroni	Spending	Exact boundary	Uncorrected
Linear path, slope 0.00	0.019	0.023	0.042	0.048	0.601
Linear path, slope 0.15	0.018	0.025	0.046	0.049	0.602
Linear path, slope 0.30	0.020	0.025	0.045	0.049	0.597
Lag-6 correlation 0.25	0.011	0.022	0.043	0.044	0.573
Lag-6 correlation 0.50	0.006	0.019	0.043	0.037	0.498
Lag-6 correlation 0.80	0.000	0.010	0.045	0.019	0.323
True SE 1.10 times published	0.047	0.062	0.086	0.109	0.733
True SE 1.25 times published	0.119	0.160	0.177	0.243	0.876
True SE 1.50 times published	0.318	0.415	0.398	0.524	0.971
True SE 1.75 times published	0.531	0.661	0.628	0.756	0.994

3.6.2 Proportional redesign effects and misdated regimes If the post-redesign path is a multiple of the pre-redesign path, then within each regime the path is still linear, with a different slope. The nuisance space can therefore contain a separate slope for each regime, and the monitor then tests only for curvature within regimes. Proposition 1 applies unchanged, because the added columns are zero on the rows before their regime begins and so the residual σ -fields remain nested. If the regime date is known to within g waves, the $2g + 1$ waves around the announced date can be given a level of their own and are absorbed; we call these guard waves. Regime slopes and guard waves can be combined; both cost information (Table 5).

3.6.3 Scale of the series If the redesign effect is multiplicative, a log transformation with delta-method standard errors s_j/y_j makes it additive; the null is then that the log series is linear, that is, exponential growth at a constant rate. The choice of scale is a substantive choice: it determines what is called curvature.

3.6.4 Mixture over the prior scale A convex combination of e-processes is an e-process. We use the equal-weight mixture of $E_n^\pm$ or $E_n^{\sigma,\pm}$ over $\tau \in \{0.0025, 0.005, 0.01, 0.02, 0.04\}$, which is valid under the same conditions as the single-scale processes and replaces the choice of τ by the choice of a grid. The grid contains the larger values at which the summer 2025 slowdown is declared (Subsection 5.4).

4 Simulation study

All simulations use the dates and design-based standard errors of the 54 legacy-wording BTOS waves, with 20,000 replications unless stated. Five rules monitor the curvature statistic $A_n/\sqrt{B_n}$ over the 52 looks from the third wave. The e-process uses $\tau = 0.01$ and $\alpha = 0.05$. The Bonferroni rule uses level $\alpha/52$ at each look and needs the horizon in advance. The spending rule uses level $\alpha/(k(k+1))$ at look k and does not. The exact boundary is a constant threshold on $|A_n|/\sqrt{B_n}$, calibrated by simulation to size 0.05 for this design; it needs the horizon and the information schedule. The uncorrected rule tests at level α at every look. The simulation code is part of the replication package.

4.1 False-declaration probability

Table 1 shows that the e-process holds its level for all three slopes at about 0.02, below the nominal 0.05, because the mixture and the two-sided split are conservative at this horizon. The Bonferroni rule is similar at 0.023 to 0.025, the spending rule uses most of the budget (0.042 to 0.046), the exact boundary is calibrated to 0.05 and realizes 0.048 to 0.049, and the uncorrected rule declares at least once in 60 percent of replications, 2.6 times on average. Over the full restart procedure under a linear path the probability of at least one declaration is 0.022 and the expected number of declarations is 0.022 (slope 0) and 0.024 (slope 0.3), within the bound of Corollary 1. Lag-6 correlation, as when a rotating panel returns, makes the e-process, the Bonferroni rule and the exact boundary more conservative over these 52 looks and leaves the spending rule unchanged; this direction is not general and may reverse for longer epochs. The weak point is the standard errors. If the true standard error is 1.25 times the published one, the e-process declares falsely in 12 percent of replications, with 1.5 times in 32 percent and with 1.75 times in 53 percent, and the other rules are no better. The e-process is not more robust than the others; it is only less sensitive because it is more conservative.

With a tolerance $\gamma_0 = 0.001$, a path with curvature $\gamma = 0.001$ is declared by the monitor with $\gamma_0 = 0$ in 97 percent of replications and by the monitor with the tolerance in 1.1 percent; with $\gamma = 0.003$ both declare in all replications.

4.2 Detection of a takeoff

Table 2 shows that the e-process is less powerful than the Bonferroni rule and the exact boundary at small effects: for a slope increase of 0.02 the probability of a declaration by the end is 0.63, against 0.78 and 0.83, and the median delay is 21 waves against 20 and 19. At larger effects the three are close (0.97 to 0.99 by the end for a slope increase of 0.05, with a median delay of 11, 10 and 10 waves). The spending rule, which like the e-process does not need the horizon, is less powerful than both at every effect size. The cost of the e-process relative to the horizon-calibrated rules is therefore a loss of power at small

Table 2. Takeoff detection when the growth rate rises by the stated amount (percentage points per wave) at the 25th of 54 waves, from a baseline growth of 0.10. Each cell gives the probability of a first declaration within 20 waves of the change, the probability of one by the end of the series, and the median delay in waves among the paths detected. The probability of a declaration before the change is 0.012 to 0.015 for the e-process and the Bonferroni rule, 0.04 for the spending rule and 0.03 for the exact boundary.

Slope increase	E-process	Bonferroni	Spending	Exact boundary
0.01	0.03 / 0.08 / 22	0.06 / 0.15 / 23	0.01 / 0.02 / 24	0.09 / 0.21 / 22
0.02	0.29 / 0.63 / 21	0.41 / 0.78 / 20	0.12 / 0.39 / 24	0.49 / 0.83 / 19
0.03	0.80 / 0.97 / 16	0.87 / 0.98 / 16	0.58 / 0.91 / 19	0.89 / 0.97 / 14
0.05	0.99 / 0.99 / 11	0.99 / 0.99 / 10	0.96 / 0.96 / 13	0.97 / 0.97 / 10

Table 3. Probability of a false declaration of the basic monitor after a redesign at the 21st wave, with 2, 5, 10 or 20 waves in the new regime, on a linear path of slope 0.25. The additive shift is 7.3 points; the multiplicative effect scales the series by 1.73; in the next two columns the shift is additive but the regime label is placed one wave before or after the true change. The last column is the pooled analysis with an additive shift. 3,000 replications (5,000 for the pooled analysis).

Waves after redesign	Additive shift	Multiplicative effect	Label one wave early	Label one wave late	Pooled analysis
2	0.010	0.012	1.000	1.000	1.000
5	0.013	0.027	1.000	1.000	1.000
10	0.017	1.000	1.000	1.000	1.000
20	0.019	1.000	1.000	1.000	1.000

Table 4. Monitor variants after a redesign, with 20 waves in the new regime (3,000 replications). The first four columns give the probability of a false declaration. Information is the final value of B_n relative to the basic monitor under the additive shift.

Monitor	Additive shift	Multiplicative effect	Label early	Label late	Information
Regime levels (baseline)	0.019	1.000	1.000	1.000	1.00
Guard waves	0.020	1.000	0.019	0.021	0.86
Regime slopes	0.014	0.012	1.000	1.000	0.07
Regime slopes and guard waves	0.013	0.011	0.011	0.016	0.05

Table 5. Information retained by the variants of Table 4 under the additive shift, relative to the basic monitor, by the number of waves in the new regime.

Waves after redesign	Guard waves	Regime slopes	Both
2	0.81	0.99	0.81
5	0.74	0.87	0.71
10	0.69	0.38	0.31
20	0.86	0.07	0.05

effects, partly from its realized size of 0.02 and partly from the prior scale, in exchange for a guarantee that does not depend on the horizon.

4.3 Redesigns

Table 3 separates what the guarantee of the basic monitor covers from what it does not. For an additive shift at the correct date it holds its level at every length of the new regime, and the pooled analysis declares in every replication. For a proportional effect, which scales the post-redesign path and so changes its slope, the monitor is fine for 2 and 5 new waves (0.012 and 0.027), because the new regime carries little information, and fails from 10 waves on. A regime label one wave early or late fails at every length. A short post-redesign run, as in the BTOS application, therefore cannot reveal whether the redesign effect is additive.

4.4 Extensions

Table 4 shows that the redesign extensions do what they are built for, at a cost that depends on the length of the new regime (Table 5). Guard waves bring the false-declaration probability of a label one wave off from 1.0 to 0.019 and 0.021 and keep 86 percent of the information with 20 new waves, and 69 to 81 percent with 2 to 10. Regime slopes bring the proportional effect from 1.0 to 0.012 but keep 99 percent of the information with 2 new waves, 38 percent with 10 and 7 percent with 20, because the basic monitor draws most of its information from curvature across the redesign, which regime slopes remove. With both, every scenario is held at 0.011 to 0.016, with 81, 71, 31 and 5 percent of the information. Regime slopes therefore protect against a hypothesis that cannot be checked at the cost of most power after a long post-redesign run, and at almost no cost after a short one.

Table 6 shows two things about the unknown-scale monitors. The first is that they hold their level when the published standard errors are too small by a common factor: 0.027 for a factor of 1.5 and 2, and 0.029 for the mixture at 1.5, where the known-scale monitor declares falsely in 31 and 71 percent of replications. They also hold it when the weights are wrong

Table 6. Unknown-scale monitors against the known-scale monitor over 54 waves (1,500 replications; 1,000 for the mixture). The first block is the probability of a false declaration when the published standard errors are too small by the stated factor; the second block when the errors are dependent or the weights are wrong; the third block is the probability of a first declaration by the end of the series when the growth rate rises by the stated amount at the 25th wave. Monte Carlo standard errors are about 0.005 for the false-declaration entries near 0.03.

Scenario	Unknown scale	Unknown scale, mixture	Known scale
True SE 1.0 times published	0.035	0.016	0.019
True SE 1.5 times published	0.027	0.029	0.311
True SE 2.0 times published	0.027	—	0.713
Lag-1 error correlation -0.3	0.001	—	0.001
Lag-1 error correlation 0.1	0.051	—	0.039
Lag-1 error correlation 0.3	0.127	—	0.113
Lag-1 error correlation 0.5	0.328	—	0.230
True SD constant at 0.15 (published weights vary)	0.043	—	0.149
Wave-specific scale errors, lognormal SD 0.3	0.028	—	0.058
Wave-specific scale errors, lognormal SD 0.5	0.023	—	0.163
Detection by end, slope increase 0.02	0.519	0.587	0.632
Detection by end, slope increase 0.03	0.957	0.963	0.972
Detection by end, slope increase 0.05	0.983	0.986	0.988
Detection by end, slope increase 0.02, true SE 1.5 times published	0.162	—	—
Detection by end, slope increase 0.03, true SE 1.5 times published	0.556	—	—
Detection by end, slope increase 0.05, true SE 1.5 times published	0.973	—	—

by wave-specific amounts (0.043 when the true standard deviation is constant at 0.15 against 0.149 for the known-scale monitor, and 0.023 to 0.028 for lognormal wave-specific errors), although the first is close to the nominal level. The cost is power when the standard errors are right: for a slope increase of 0.02 the probability of a declaration by the end is 0.52 for the unknown scale, 0.59 for the unknown scale with the mixture and 0.63 for the known scale; at 0.03 and 0.05 the three are within 0.02. With standard errors 1.5 times too small the unknown-scale monitor detects a slope increase of 0.03 in 56 percent of replications and of 0.05 in 97 percent.

The second is that dependence is not covered, and this includes the unknown-scale monitors. With a first-order error correlation of 0.1 the unknown-scale monitor is already at the nominal level (0.051), and with 0.3 and 0.5 it declares falsely in 13 and 33 percent of replications, the known-scale monitor in 11 and 23 percent. Negative correlation makes both conservative (0.001). Positive dependence between adjacent waves is what wave-level roughness of the latent level produces, which Subsection 5.7 suggests is present in the BTOS series; the recommended monitor is therefore valid for the application only if that roughness is small. The mixture over the prior scale for the known-scale monitor has a false-declaration probability of 0.017 to 0.019 and detects a slope increase of 0.02 by the end in 68 percent of replications, against 63 percent for a single scale, with the same median delay. On a log scale, with exponential growth and a relative standard error of 3 percent, a multiplicative redesign effect is held at 0.009 to 0.021 for 2 to 20 new waves, and an additive effect on the level scale fails from 10 waves (0.97 and 1.0).

5 Application: business use of artificial intelligence

5.1 Data

The BTOS is an experimental survey of about 1.2 million nonfarm employer businesses divided into six rotating panels; each panel reports every twelve weeks and collection occurs every two weeks (U.S. Census Bureau 2025). Two questions on artificial intelligence have been asked since the collection period beginning 11 September 2023 (Bonney et al. 2024). Until the collection period ending in October 2025 the question on current use read “in producing goods or services”; from the collection beginning 17 November 2025 it read “in any of its business functions” (U.S. Census Bureau 2026), and question wording is known to matter for measured adoption (Bick 2026). The 15 December 2025 file of the Census Bureau removed the legacy-wording rows, so we combine the archived vintages of 8 and 15 December 2025 (Goldschlag 2025). The data set has 56 published national waves: 54 legacy-wording waves with collection ending 24 September 2023 to 5 October 2025, and two revised-wording waves ending 30 November and 14 December 2025. No estimate was published for the three intervening periods, which appear as a gap in the time index. National standard errors have a median of 0.15 points (range 0.08 to 0.23). Estimates and standard errors are also available for seven employment size classes. The series ends in December 2025, the end of the last full year. The data are an archived final series and not the vintages that a live user would have seen, so the dates below describe the series as it now stands.

Under the legacy wording the national estimate rose from 3.7 percent to 4.9 percent by the end of 2023, fell to 4.3 percent in March 2024, and then rose to 10.0 percent in October 2025. Under the revised wording it was 17.3 and 17.2 percent.

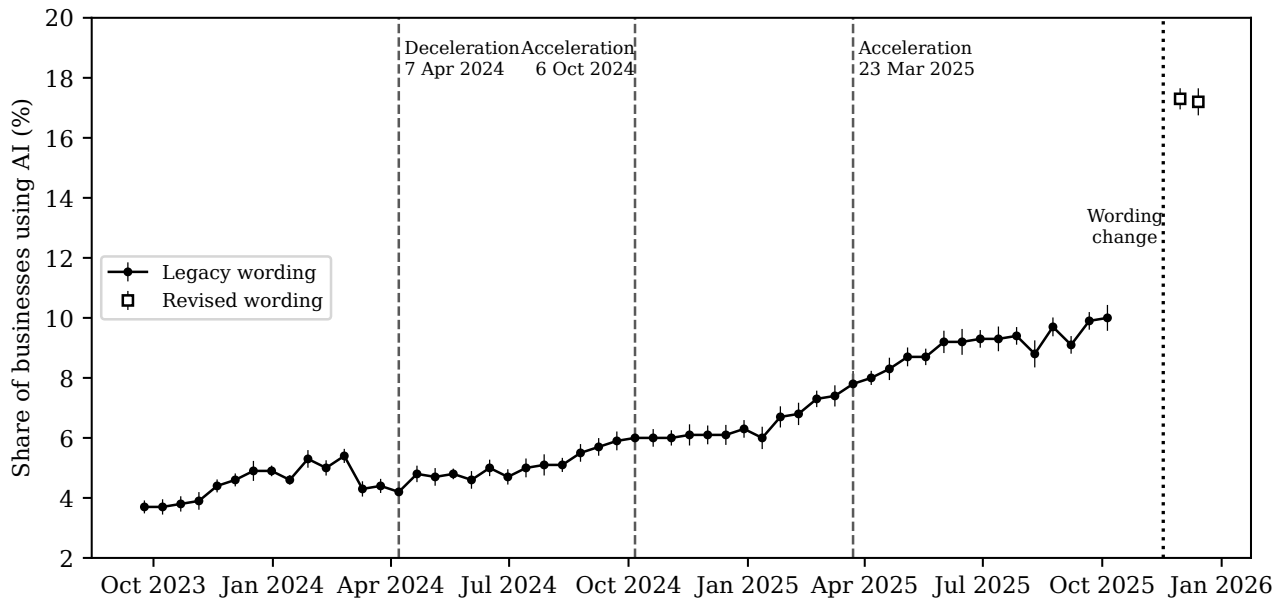


Figure 1. National estimate of the share of businesses using artificial intelligence, September 2023 to December 2025, with 95 percent intervals. Dashed vertical lines mark the declarations of the primary monitor. The dotted vertical line marks the wording change of 17 November 2025. Circles: legacy wording; open squares: revised wording.

Table 7. Epochs of the primary monitor on the national series (unknown scale, mixture over the prior scale, $\alpha = 0.05$). Waves is the number of waves in the epoch up to the declaration. Curvature is A_n/B_n at declaration with a 95 percent half-width using the scale estimated within the epoch. For the last epoch, which ends without a declaration, E is the maximum; its anytime-valid p-value is 0.11.

Epoch start	Waves	Declaration	Date	Curvature (pp per wave ²)	E
24 Sep 2023	15	Deceleration	7 Apr 2024	-0.0390 ± 0.0262	131.3
21 Apr 2024	13	Acceleration	6 Oct 2024	0.0276 ± 0.0196	70.5
20 Oct 2024	12	Acceleration	23 Mar 2025	0.0417 ± 0.0302	55.0
6 Apr 2025	16	None	—	—	18.0

5.2 Dispersion of the published standard errors

The standardized wave-to-wave changes within the legacy regime have a lag-one autocorrelation of -0.49 , as expected for independent errors around a slowly moving path. The standardized second differences, which remove a local linear trend, should have a standard deviation near one if the path is smooth and the errors are as published; they have 1.76 (1.55 by a robust estimate based on the median absolute deviation). There are two readings. The published standard errors may be too small by a factor of about 1.5. Or the latent path may be rougher than a smooth curve, for example because the level moves from wave to wave through nonsampling error or panel effects. The published figures cannot separate them, and Subsection 5.7 finds that a state-space model with a random-walk level favours the second. Under either reading the null of the monitor, exact linearity with independent sampling errors, is false. Under the first the unknown-scale monitor is the appropriate one; under the second the errors are positively dependent, which that monitor does not cover (Table 6).

5.3 Primary analysis

The monitor we treat as primary is the unknown-scale monitor with the mixture over the prior scale, because it needs the fewest assumptions among the versions with a proof. This designation was made after the exploratory analyses reported in this section, for that reason and not because of its results; every other version is a sensitivity analysis, and agreement between versions on the same data is not independent corroboration.

Figure 1 and Table 7 give the results. The first epoch begins on 24 September 2023 and the deceleration monitor declares on 7 April 2024, after 15 waves. The trigger is a single wave: on 10 March 2024 the estimate falls from 5.4 to 4.3 percent, a change of -6.2 combined standard errors and the largest within-regime change in the series, and it stays near 4.4 afterwards, so this is a persistent drop and not a flat spell. The cross-strata check finds three of the seven size classes beyond their own 95th percentile with the same sign at that wave, the only legacy wave with three or more, against seven of seven at the wording change. The evidence does not distinguish a genuine fall in business use from an unannounced collection or processing change. The second epoch begins on 21 April 2024 and declares an acceleration on 6 October 2024, the third begins on 20 October 2024 and declares an acceleration on 23 March 2025, and the fourth begins on 6 April 2025 and runs through the wording change; its deceleration monitor peaks at 18.0 on 7 September 2025, an anytime-valid p-value of 0.11,

Table 8. Declarations under variants of the monitor on the national series. Dec.: deceleration; acc.: acceleration. For the epochs that end without a declaration the maximum E in the last epoch is given.

Variant	Declarations
Primary monitor	7 Apr 2024 (dec.); 6 Oct 2024 (acc.); 23 Mar 2025 (acc.) (last epoch peak 18.0)
Primary, wave of 10 March 2024 absorbed by its own level	7 Apr 2024 (dec.); 6 Oct 2024 (acc.); 23 Mar 2025 (acc.)
Primary, level shift from 10 March 2024	none (last epoch peak 19.8)
Primary, monitoring starts 3 waves later	9 Feb 2025 (acc.); 7 Sep 2025 (dec.)
Primary, monitoring starts 6 waves later	22 Sep 2024 (acc.); 9 Mar 2025 (acc.)
Unknown scale, single $\tau = 0.005$	7 Apr 2024 (dec.); 6 Apr 2025 (acc.) (last epoch peak 2.0)
Unknown scale, single $\tau = 0.01$	7 Apr 2024 (dec.); 6 Oct 2024 (acc.); 6 Apr 2025 (acc.) (last epoch peak 3.6)
Unknown scale, single $\tau = 0.02$	7 Apr 2024 (dec.); 22 Sep 2024 (acc.); 9 Mar 2025 (acc.); 7 Sep 2025 (dec.) (last epoch peak 1.0)
Unknown scale, single $\tau = 0.04$	7 Apr 2024 (dec.); 22 Sep 2024 (acc.); 23 Feb 2025 (acc.); 27 Jul 2025 (dec.) (last epoch peak 1.0)
Primary, regime slopes and guard waves	7 Apr 2024 (dec.); 6 Oct 2024 (acc.); 23 Mar 2025 (acc.)
Unknown scale on the log scale, $\tau = 0.002$	7 Apr 2024 (dec.); 4 May 2025 (acc.)
Basic monitor (known scale, $\tau = 0.01$)	10 Mar 2024 (dec.); 6 Apr 2025 (acc.) (last epoch peak 19.0)
Basic monitor, tolerance $\gamma_0 = 0.003$	10 Mar 2024 (dec.); 4 May 2025 (acc.)
Basic monitor, tolerance $\gamma_0 = 0.004$	10 Mar 2024 (dec.); 1 Jun 2025 (acc.)
Basic monitor, tolerance $\gamma_0 = 0.005$	10 Mar 2024 (dec.)
Basic monitor, SE multiplied by 1.55	10 Mar 2024 (dec.); 6 Apr 2025 (acc.)

Table 9. Employment size classes at the wording change. Ratio is the first revised estimate over the last legacy estimate; jump is the standardized change; the 95th percentile is that of the class's standardized wave-to-wave changes within the legacy regime. The last column gives the date on which the pooled basic monitor declares an acceleration at the wording change, where it does.

Size class	Last legacy (%)	First revised (%)	Ratio	Jump $ z $	95th pct. within	Pooled declares at wording change
1–4	11.0	17.0	1.55	13.2	2.77	30 Nov 2025
5–9	8.6	15.5	1.80	8.7	2.85	30 Nov 2025
10–19	8.1	17.8	2.20	10.6	2.37	30 Nov 2025
20–49	7.0	18.7	2.67	13.8	1.96	30 Nov 2025
50–99	9.7	20.7	2.13	9.7	2.29	30 Nov 2025
100–249	8.8	23.7	2.69	8.6	1.74	no
250+	13.0	30.5	2.35	6.1	2.31	no

and does not declare. The two revised-wording waves are retained with a level of their own. The curvature estimates are large for such short epochs, with wide intervals, and the scale estimated within the first epoch is 4.1 times the published standard error because of the drop.

5.4 Sensitivity

Table 8 is the main caution of this paper. The statements that survive every variant are that the series is not a straight line, and that the estimate drops in March 2024. Little else survives. If the March 2024 drop is treated as a level shift that persists, the primary monitor makes no declaration at all, with peak evidence of 19.8, and the basic monitor changes to three decelerations in 2024 and 2025 with no acceleration. If only the wave itself is absorbed, the first declaration is unchanged. The sequence depends on where monitoring starts: beginning three waves later gives an acceleration on 9 February 2025 and a deceleration on 7 September 2025 ($E = 207$), the summer 2025 slowdown, which beginning on the original date does not. It depends on the prior scale: with a single τ of 0.02 or more the unknown-scale monitor declares the slowdown ($E = 64$ on 7 September 2025 at 0.02; $E = 163$ on 27 July 2025 at 0.04), and with 0.005 it does not declare the second 2024 acceleration. It depends on the tolerance: for the basic monitor the acceleration of spring 2025 moves to May 2025 at $\gamma_0 = 0.003$ and disappears at 0.005, about its estimated curvature. Regime slopes and guard waves change nothing before the wording change, because the two revised waves are absorbed. On the log scale the first declaration is on the same date and the acceleration is in May 2025. The summer 2025 slowdown is therefore best described as inconclusive, and the dates of the other declarations as uncertain by months.

5.5 The wording change

Between the last legacy wave (10.0 percent, 5 October 2025) and the first revised wave (17.3 percent, 30 November 2025) the estimate rose by 7.3 points, which is 25.7 combined standard errors, against a within-regime median absolute standardized change of 0.94 and a maximum of 6.2. The regime-aware monitors book no evidence from it, by construction, and with two waves in the new regime this says little beyond that the transition wave is absorbed. The pooled analysis declares an acceleration on 30 November 2025 with an e -value of order 10^{123} , and a repeated curvature test applied to the spliced series has a statistic of 36.4 in magnitude at that wave.

Table 10. State-space models for the legacy and revised waves with the published standard errors multiplied by an estimated common factor (scale) and a level intervention at the wording change, fitted by maximum likelihood; slopes are in points per wave, with 95 percent half-widths. The log-likelihoods are comparable across rows (diffuse initial states). Slope var. and level var. are the innovation variances per wave.

Model	Log-lik.	Scale	Slope var.	Level var.	Slope Apr 2025	Slope Jul 2025	Jul minus Apr
Smooth trend (slope disturbance only)	15.2	1.65	5.7e-03	0.000	0.28 ± 0.12	0.05 ± 0.12	-0.22 ± 0.17
Local linear trend (level and slope disturbances)	19.4	1.12	3.8e-05	0.058	0.12 ± 0.07	0.12 ± 0.08	-0.00 ± 0.03
Random-walk level, constant slope	19.4	1.11	0.0e+00	0.060	0.12 ± 0.07	0.12 ± 0.07	0.00 ± 0.00

Table 9 shows that all seven size classes jump in the same wave and in the same direction, by 6.1 to 13.8 standard errors and by 2.6 to 7.0 times their own 95th percentile (1.74 to 2.85). The ratio of the first revised to the last legacy estimate, 1.55 to 2.69, and the difference, 6.0 to 17.5 points, both rise with the size class, so the effect is neither a constant nor a common proportion, and it is 1.73 at the national level. With two revised waves we cannot test whether the effect on any one series is additive, and Table 3 shows that this matters once the new regime is longer.

5.6 Comparison with repeated testing

The curvature z statistic of the basic, known-scale analysis was applied at each wave with the same restart rule, with regime columns and without. The uncorrected test declares nine times, the first on 19 November 2023, and also on 14 December 2025 with regime columns and on 30 November 2025 without. The Bonferroni rule, using the remaining length of the series as its horizon, declares on 10 March 2024, 6 April 2025, 10 August 2025 and 21 September 2025; the last two are the summer 2025 slowdown, which the basic monitor with $\tau = 0.01$ does not declare, as expected from the loss of power in Table 2. The spending rule declares six times, with the first and the last two on the same dates as the Bonferroni rule. A moving-average rule, which crosses when the sign of the 3-wave mean minus the 8-wave mean changes, crosses four times (24 March 2024, 19 May 2024, 7 September 2025, 21 September 2025). Without regime columns the Bonferroni and spending rules add a declaration on 14 December 2025 at the revised-wording waves. The e-process is therefore not distinguished on the real data by what it finds; its advantage is the guarantee, at a cost in power.

5.7 Comparison with a state-space model

The monitor tests a hypothesis; a user who wants the size of a change fits a model. We fitted three Gaussian state-space models (Table 10). The smooth-trend model, in which the slope follows a random walk and the level has no disturbance, estimates a scale factor of 1.65 on the published standard errors and a slope of 0.28 ± 0.12 points per wave in April 2025 against 0.05 ± 0.12 in July 2025, a difference of -0.22 ± 0.17 . The standard local linear trend model, which adds a level disturbance, has a log-likelihood 4.2 higher, a scale factor of 1.12, a slope variance of practically zero and a level variance of 0.058, and its smoothed slope is 0.11 to 0.12 points per wave at all six dates we looked at. A random-walk level with a constant slope fits as well as the local linear trend (log-likelihoods 19.4 and 19.4), and with the scale fixed at one it fits almost as well (19.3). In this model the published standard errors are about right, there is no evidence of a change in the growth rate, and the departure from a straight line is wave-level roughness of the level. This is the second reading in Subsection 5.2. The dates for the smoothed slopes were taken from the declarations of the monitors and from the level-shift analysis, so they are not independent of the data.

The curvature monitors reject exact linearity with independent sampling errors. The state-space comparison shows that this rejection does not by itself say that the growth rate changed: a constant growth rate with a random-walk level produces it too. The declarations should therefore be read as departures from a straight line with white sampling noise, and as candidates for the changes in growth that the first model suggests, and not as established changes. The standardized one-step innovations, from 46 waves and against a band of about ± 0.30 , have autocorrelations of -0.06 , 0.21 and -0.43 at lags 1, 2 and 3 for the smooth-trend model and -0.24 , 0.43 and -0.32 for the local linear trend, so some short-lag structure remains in both, which we attribute in part to the March 2024 wave and have not investigated further. At lags 6 and 12 they are 0.14 and -0.10 and 0.17 and -0.10 : there is no evidence of dependence at the lag at which a panel returns.

6 Discussion

6.1 Implications for agencies and users

Agencies that announce a redesign should publish the regime label with the series, and, for a redesign whose effect is unlikely to be additive, a parallel run or a bridge; the monitor needs the label, and with the guard an error of one wave costs between 14 and 31 percent of the information, depending on the length of the new regime (Table 5). Users who monitor a series should report the epoch, the estimated curvature and its standard error, and the sensitivity to the scale of the standard errors and to the starting date, in addition to the e-value. The e-process is a complement to a state-space model with an intervention, which estimates the slope and its change directly; it is useful where a user wants a stopping rule with a stated error probability that holds however often the series is inspected, and it should be read as a test of exact linearity with independent sampling errors.

6.2 Limitations

The extensions remove the assumptions of known scale, additive redesign effects, exact regime dates and a fixed prior scale, but not all assumptions. The unknown scale is a common factor: standard errors that are wrong by wave-specific amounts, which the published ones may be, since the national ones range from 0.08 to 0.23 points, are covered only approximately (Table 6), and the proof still takes the weights as fixed. Positive dependence between adjacent waves, which wave-level roughness of the level produces, is not covered: with a first-order correlation of 0.3 the unknown-scale monitor declares falsely in 13 percent of replications. The null is a statement about shape, and a declaration says that the path is not a straight line, not why: in the application a random-walk level with constant growth explains the data as well as the declarations suggest changes of growth. The first declaration is driven by one large drop that may be a measurement event, and the cross-strata check is a screening rule. Declarations depend on where the epoch starts, on the prior scale and on the tolerance, and the dates are crossing times. The corollaries on restarts are proved under linearity throughout, which fails for the series; no guarantee covers the sequence of declarations actually reported. The primary monitor was designated after the exploratory analysis. The data are an archived final series ending in December 2025, the post-redesign run is two waves long, and nonresponse bias that drifts over time is absorbed into the latent path.

7 Conclusion

A published survey series can be monitored for departures from a straight line with error control that holds under continuous inspection, for any unknown level and slope and for redesign shifts at announced dates. The basic monitor needs the published standard errors to be right, the shifts to be additive and the dates to be exact; the extensions remove each of these in turn, at a cost in information that we quantify, and none covers dependent errors. Applied to the BTOS series on business use of artificial intelligence, the recommended monitor declares a deceleration in April 2024, triggered by one large drop, and accelerations in October 2024 and March 2025. No declaration remains if the drop is treated as a level shift, the summer 2025 slowdown is declared or not depending on the starting date and the prior scale, and a state-space model with a random-walk level and a constant growth rate fits as well as the declarations suggest changes of growth. The monitor is a screening tool for departures from a straight line, and the size of any change in growth should be estimated with a model.

A Proof of Proposition 1 and Corollary 1

Fix an epoch and write $\mathcal{X}_n = \text{span}(D_n) \subset \mathbb{R}^n$. Under model (1) the vector $y_{1:n}$ equals $D_n c + \gamma q_{1:n} + e_{1:n}$ for some $c \in \mathbb{R}^p$ and $e_{1:n} \sim \mathcal{N}(0, W_n^{-1})$. Let G_n be the σ -field generated by $R_n = M_n y_{1:n}$, that is, by $y_{1:n}$ modulo $\mathcal{X}_n$. The columns of D_{n+1} restricted to the first n rows span $\mathcal{X}_n$, with a new regime column being zero on earlier rows, so $y_{1:n} + \mathcal{X}_n$ is a function of $y_{1:n+1} + \mathcal{X}_{n+1}$ and (G_n) is a filtration. Moreover $B_{n+1} \geq B_n$: the residual of q on the larger design restricted to the first n rows is no larger in the W -norm than the residual on D_n , and the new row adds a nonnegative term.

In the inner product $\langle u, v \rangle_W = u^\top W_n v$, M_n is the orthogonal projector onto $\mathcal{X}_n^\perp$. The vector R_n is a standard Gaussian on $\mathcal{X}_n^\perp$ with mean $\gamma M_n q$, and its law does not depend on c . Write $u = \delta - \gamma_0$ for a candidate curvature $\delta > \gamma_0$ and $A_n^+ = A_n - \gamma_0 B_n$, where $\langle M_n q, R_n \rangle_W = A_n$ and $\langle M_n q, M_n q \rangle_W = B_n$. The density ratio of R_n under curvature δ against curvature γ_0 is

$$L_n(u) = \exp(u A_n^+ - \frac{1}{2} u^2 B_n).$$

For $\gamma = \gamma_0$, (L_n) is the likelihood ratio of the observation R_n of fixed law on the increasing sequence G_n and is a nonnegative martingale with initial value one. Mixing over $u > 0$ with the half-normal law keeps this and gives E_n^+ by evaluating the Gaussian integral,

$$\int_0^\infty e^{u A - u^2 B/2} \frac{2}{\tau \sqrt{2\pi}} e^{-u^2/(2\tau^2)} du = \frac{2}{\tau \sqrt{a}} \Phi\left(\frac{A}{\sqrt{a}}\right) e^{A^2/(2a)}, \quad a = B + \tau^{-2}.$$

For $\gamma < \gamma_0$, conditionally on G_{n-1} the increment $A_n^+ - A_{n-1}^+$ is normal with mean $(\gamma - \gamma_0)(B_n - B_{n-1})$ and variance $B_n - B_{n-1}$ (the innovation of a Gaussian regression in sequentially orthogonalized form). Hence $\mathbb{E}[L_n/L_{n-1} \mid G_{n-1}] = \exp(u(\gamma - \gamma_0)(B_n - B_{n-1})) \leq 1$ for $u > 0$, because $B_n - B_{n-1} \geq 0$, and the mixture is a supermartingale. Ville's inequality gives $\mathbb{P}(\sup_n E_n^+ \geq 2/\alpha) \leq \alpha/2$ for all c . The case of E^- is the same with the sign of A_n reversed. This proves Proposition 1.

For Corollary 1, condition on the history up to the stopping time at which the epoch starts. Given the history, the future waves obey the model and the latent path is a fixed function. If the path is linear on the first n^* waves of the epoch, a declaration at $n \leq n^*$ is an event of the type bounded in Proposition 1, applied to the model whose path is the linear extension, which has the same law of the data up to n^* . If the path is linear throughout, each epoch ends in a declaration with conditional probability at most α , so $\mathbb{P}(N \geq m) \leq \alpha^m$ and $\mathbb{E}[N] = \sum_{m \geq 1} \mathbb{P}(N \geq m) \leq \alpha/(1 - \alpha)$.

For Proposition 2, let $\mathcal{X}_n$ be as above and write $R_n = M_n y_{1:n}$. Under $\gamma = 0$ the law of R_n is $\mathcal{N}(0, \kappa^2 I)$ on $\mathcal{X}_n^\perp$ in the W -geometry, for every c . The group of location shifts in $\mathcal{X}_n$ and positive scalings acts on the data; a maximal invariant is the direction $u_n = R_n / \|R_n\|$, whose law under $\gamma = 0$ is uniform on the unit sphere of dimension $\nu_n - 1$ for every c and κ . Under curvature γ the vector R_n is $\mathcal{N}(\gamma m_n, \kappa^2 I)$ with $m_n = M_n q$ and $\|m_n\|^2 = B_n$, and the density of u_n depends on (γ, κ) only through $\delta = \gamma/\kappa$. In polar coordinates $R_n = x u_n$ the density of u_n relative to the uniform law is proportional

to $e^{-\delta^2 B_n/2} \int_0^\infty x^{\nu_n-1} e^{-x^2/2+x\delta\langle u_n, m_n \rangle} dx$, with $\langle u_n, m_n \rangle = A_n/\sqrt{\text{RSS}_n} = z_n$, which is $L_n(\delta)$ in (5). The σ -fields of u_n are nested: R_n is a linear function of R_{n+1} , and hence u_n is a function of u_{n+1} whenever $R_n \neq 0$, which holds almost surely. So $L_n(\delta)$ is a likelihood ratio of increasing observations of fixed null law, a nonnegative martingale with mean one, and so is its mixture $E_n^{\sigma,+}$; Ville's inequality gives the bound. The monitor for deceleration is the same with $\delta < 0$. Before monitoring starts, that is, while $\nu_n < 2$ or $B_n = 0$, the σ -field is trivial and the process is constant at one, so the martingale property is not affected. The unit mean can be checked exactly: given B_n and ν_n , the cosine ζ between u_n and m_n has density proportional to $(1 - \zeta^2)^{(\nu_n-3)/2}$ on $(-1, 1)$ and $z_n = \zeta\sqrt{B_n}$, so the mean of the computed $E_n^{\sigma,+}$ can be integrated over ζ ; it is between 1.000 and 1.003 for B_n from 50 to 10^6 and $\nu_n \in \{3, 10, 40\}$ (results/extensions.json).

Regime slopes and guard waves enlarge D_n by columns that, like the regime levels, are zero on rows before the regime begins (for regime slopes) or are unit vectors on single rows (for guard waves), so the nesting argument and Proposition 1 apply unchanged. The restart argument of Corollary 1 carries over to every monitor in this paper, since it uses only the validity of the monitor within an epoch.

Parts of the argument for Proposition 1 have been checked with the Lean 4 proof assistant and Mathlib (the directory lean/ of the replication package): the identity between the likelihood ratio exponent and $\gamma A_n - \gamma^2 B_n/2$ for orthogonal projections, the monotonicity of the information, the supermartingale step for conditionally Gaussian increments, the supermartingale property of mixtures, Ville's inequality, the closed form of the half-normal mixture, and their combination into the time-uniform bound. The innovation representation of the regression score (conditionally Gaussian increments given the residual σ -field), the density ratio of the Gaussian residual vector, Proposition 2 and Corollary 1 are not formalized; the first enters the Lean statement as a hypothesis.

B Computation

For each epoch, at each wave n with $B_n > 0$ the weighted regression of y and q on regime indicators and t gives the residuals; A_n and B_n follow from (3), and $E_n^\pm$ is evaluated on the logarithmic scale. If $\max(E_n^+, E_n^-) \geq 40$ a declaration is recorded and the next epoch starts at wave $n + 1$. The unknown-scale e-value is computed by quadrature over δ on a grid of 60 equally spaced points from zero to six prior scales and 29 points around the maximum of the integrand, with the radial integral evaluated on an 80-point grid around its mode, all on the logarithmic scale. All results were produced by the scripts in the replication package, from the data of Subsection 5.1.

Declaration of Conflicting Interests

The Author declares that there is no conflict of interest.

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Data Availability

All data are public statistics of the U.S. Census Bureau (Business Trends and Outlook Survey, https://www.census.gov/hfp/btos/data_downloads). The analysis data set and the code that reproduces every number, table and figure are provided with the submission.

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